

Problem Set 15 Answers

1a. Since X does not take values in a range of integers starting from 0 to some finite value n , then X cannot be a Binomial random variable. In particular, in this question, X has value 2 or larger.

1b. Since the t-shirt draws are dependent (without replacement), then X is not a Binomial random variable. Another way to see this is that X does not take values in a range of integers starting from 0 to some finite value n , so X cannot be a Binomial random variable. In particular, in this question, X has value 1 or larger.

1c. Yes, X is a Binomial random variable with $n = 5$ and $p = 1/3$, since the 5 days' behaviors are independent, and since the probability of being sunny is the same on each day.

2a. Since X and Y are independent, we have $P(X \neq 0 \text{ \& } Y \neq 0) = P(X \neq 0)P(Y \neq 0)$. We have $P(X \neq 0) = 1 - P(X = 0) = 1 - (1/3)^3 = 1 - 1/27 = 26/27$. Similarly, we have $P(Y \neq 0) = 26/27$, and we conclude that $P(X \neq 0 \text{ \& } Y \neq 0) = (26/27)^2$.

2b. We have $P(X_j \neq 0) = 26/27$ for each j , and the random variables are independent, so $P(X_1 \neq 0 \text{ \& } X_2 \neq 0 \text{ \& } \cdots \text{ \& } X_m \neq 0) = (26/27)^m$.

3a. Yes, X is a Binomial random variable with $n = 3$ and $p = 1/6$, since the 3 bears' colors are independent, and since the probability of being green is the same for each bear.

3b. Yes, $X + Y$ is a Binomial random variable with $n = 3$ and $p = 2/6 = 1/3$, since the 3 bears' colors are independent, and since the probability of being green or yellow is the same for each bear.

4. We compute

$$P(W = -3) = P(Y = 3, X = 0) = (1/6)^3 = 1/216$$

$$P(W = -2) = P(Y = 2, X = 0) = \binom{3}{2}(1/6)^2(4/6)^1 = 1/18$$

$$P(W = -1) = P(Y = 1, X = 0) + P(Y = 2, X = 1) = \binom{3}{1}(1/6)^1(4/6)^2 + \binom{3}{2}(1/6)^2(1/6)^1 = 17/72$$

$$P(W = 0) = P(Y = 0, X = 0) + P(Y = 1, X = 1) = (4/6)^3 + \binom{3}{1}\binom{2}{1}(1/6)^1(1/6)^1(4/6) = 11/27$$

$$\text{By symmetry: } P(W = 1) = 17/72; P(W = 2) = 1/18; P(W = 3) = 1/216$$

5a. We know that $X + Y$ is a Binomial random variable with $n = 3$ and $p = 2/6 = 1/3$, so $P(X + Y > 0) = 1 - P(X + Y = 0) = 1 - (2/3)^3 = 1 - 8/27 = 19/27$.

5b. We have $P(X - Y > 0) = P(W = 1) + P(W = 2) + P(W = 3) = 17/72 + 1/18 + 1/216 = 8/27$.